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Machine learning loan default prediction

1 Introduction

In the current financial system, banks and lenders should ensure that a borrower is worth repaying a loan or not. Loss of money can be experienced by the institution in case a borrower defaults. It is due to this that the prediction of loan default is very crucial.

In this project, machine learning is applied to analyze the information about borrowers, including their income, employment status, credit record, and lending details. The aim is to determine the key factors influencing loan default and develop a model that can specify the possibility of a loan default by a borrower.

1.1 Problem Formulation

The conventional credit rating systems tend to follow a few financial pointers and rigid regulations. Nevertheless, the behavior of borrowers and the financial risk are subject to various interacting variables including income level, credit history, employment status and debt to income ratio.

In this project, data is used to derive the results by:

1. Examining the characteristic of borrowers that affect loan default.
2. Incorporating predictive models in classification of loan default.

Research Question

What are the risk factors of loan default and can machine learning models provide an accurate prediction of if a borrower will default on a loan?

1.2 Data Acquisition

The data utilized in this research was obtained on Kaggle and can be accessed publicly. It contains personal information of the borrower, financial, loan and repayment outcomes of a simulated lending environment. The data contains 18 variables, and the variable of interest is the default that demonstrates a case when the borrower did not repay the loan (1 = Default, 0 = No Default). This information can be applied in the analysis of the behavior of borrowers and forecasting the risk of defaulting loans.

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Education	EmploymentType
0	I38PQUQS96	56	85994	50587	520	80	4	15.23	36	0.44	Bachelor's	Full-time
1	HPSK72WA7R	69	50432	124440	458	15	1	4.81	60	0.68	Master's	Full-time
2	C1OZ6DPJ8Y	46	84208	129188	451	26	3	21.17	24	0.31	Master's	Unemployed
3	V2KKSFM3UN	32	31713	44799	743	0	3	7.07	24	0.23	High School	Full-time
4	EY08JDHTZP	60	20437	9139	633	8	4	6.51	48	0.73	Bachelor's	Unemployed

2 Data Analysis

2.1 Clean and Preprocess Data

A number of preprocessing steps were carried out to remove data quality and consistency errors.

1. **Remove Unnecessary Columns**

The dataset included a column of uniqueness to identify loans (LoanID), which was eliminated since it is not a predictive aspect.

2. **Handle Missing Values**

The data set was verified on whether it contained any missing values on all variables. There were no any significant missing values. The possible missing values were substituted with values that were either correct or corrective of the models.

3. **Categorical Variables: encode.**

The categorical variables were coded based on label encoding of categorical variables Education, EmploymentType, MaritalStatus and LoanPurposeto encode into numerical values that could be used in machine learning algorithms.

4. **Remove Duplicates**

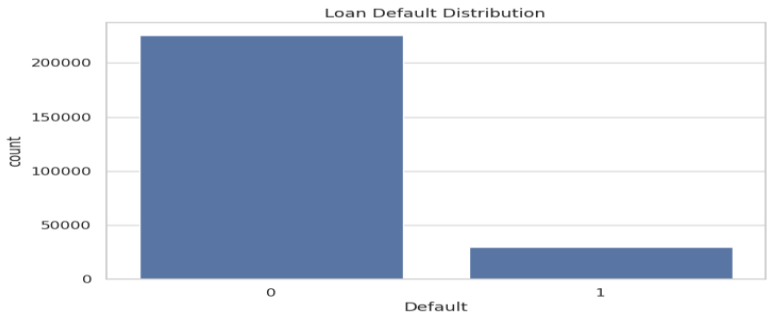
To ensure the integrity of datasets, duplicate records were verified and eliminated. The processed dataset was now prepared to undergo exploratory data analysis and machine learning model.

2.2 Exploratory Data Analysis (EDA)

To establish patterns and the relationship between loan default and the characteristics of borrowers, EDA was carried out.

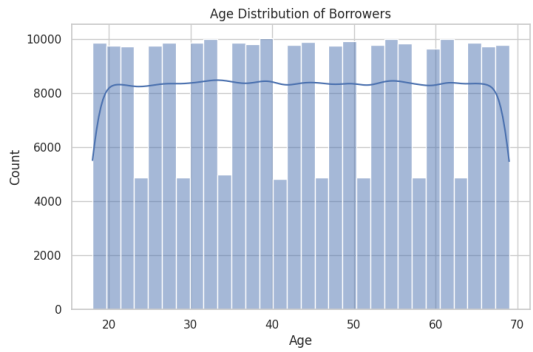
1. **Loan Default Distribution**

The target variable distribution shows that cases of non-default are more common than the cases of default, which indicates the existence of class imbalance in the dataset.



(Figure 1: Loan Default Distribution)

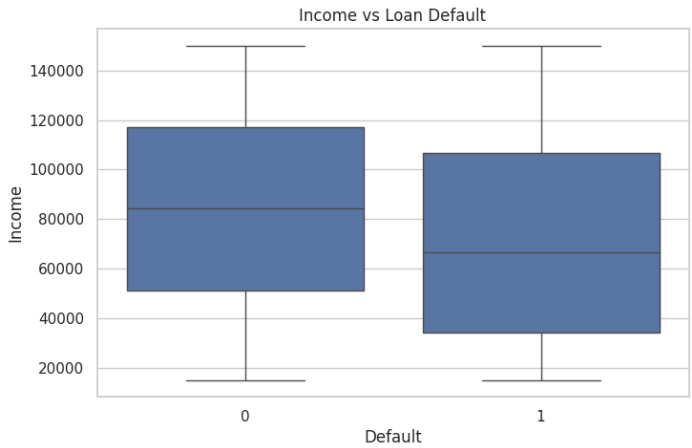
2. Age Distribution



(Figure 2: Age Distribution Histogram)

3. Income vs Loan Default

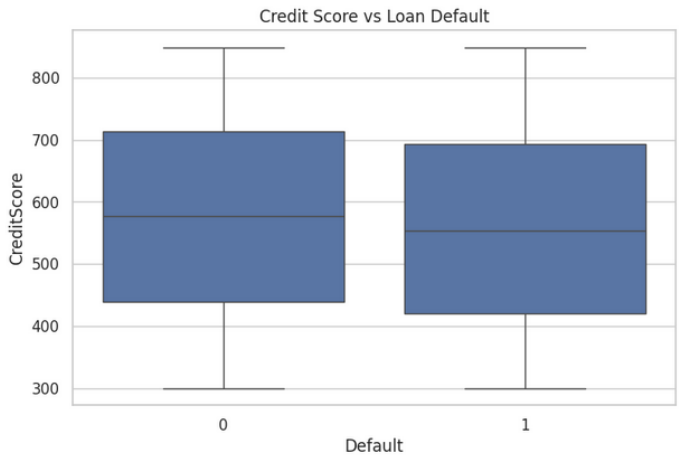
Borrowers who have lower incomes have a somewhat more high chance of defaulting as indicated by a box-plot analysis.



(Figure 3: Income vs Default Boxplot)

4. Credit Score vs Loan Default

Credit score shows only a weak relationship with loan default in this dataset.



(Figure 4: Credit Score Distribution)

5. Correlation Analysis

The correlation heatmap was applied to research the correlation between the borrowers characteristics and default on loans. The relationships with most of the variables were weak with the target variable so there is no one factor that predicts default strongly. Age was least correlated with negative (-0.17) and it indicated that old borrowers are a little less likely to default. The most positive correlation was found between interest rate (0.13) indicating the possibility that higher interest loans had a slightly higher risk of defaulting. There were also small negative amounts of income and months employed, which indicates that financially stable borrowers tend to default less. In sum, such poor correlations demonstrate that multiple factors have to be combined in the machine learning models.

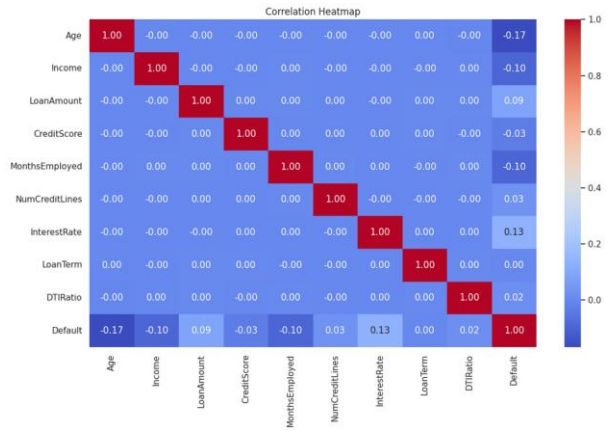


Figure 5: Correlation Heatmap

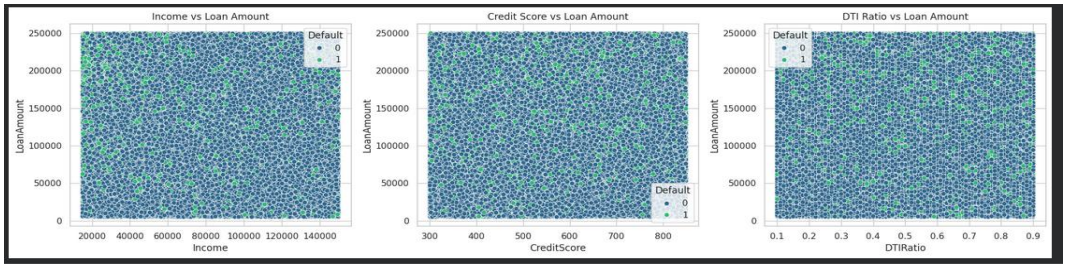


Figure 6: Scatter Plot

3 ML /AI Methods

Age, Income, Loan Amount, Credit Score, Number of credit lines, interest rate, months employed, loan term, and DTI ratio were the predictor variables which were used in this study. These characteristics have been taken as they can influence defaulting of a borrower in a loan. The predictor variable was Default, indicating the default of the borrower to repay the loan (1= Default, 0= No Default). The data consisted of numbers and coded categorical data, which is why the machine learning classification model was applicable. To evaluate the model, the data was divided into 70 percent training and 30 percent testing data.

```
# Select features and target
X = cleaned_df[['Age','Income','LoanAmount','CreditScore',
                'NumCreditLines','InterestRate','MonthsEmployed',
                'LoanTerm','DTIRatio']]

y = cleaned_df['Default']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=88
)
```

Figure 7: Feature Selection

3.1 Logistic Regression

A baseline classifier that was implemented to predict the default of a loan was the Logistic Regression model. The model attained the accuracy of 0.8836 which means that it predicted correctly some 88.36 percent. The recall score was however not very high (0.0256) which implies that the model was not good at correctly recognizing borrowers who actually defaulted.

The value of 0.0489 of the F1-score indicates that the model is not as effective in identifying the minority category (loan defaults). All in all, Logistic Regression is an easy to understand and interpret model but one with low-complexity in terms of reflecting the complex relationships in the data.

```python
# Logistic Regression
lr_model = LogisticRegression(max_iter=1000, random_state=88)
lr_model.fit(X_train, y_train)
lr_pred = lr_model.predict(X_test)

print("\nLogistic Regression Performance:")
print("Accuracy:", accuracy_score(y_test, lr_pred))
print("Precision:", precision_score(y_test, lr_pred))
print("Recall:", recall_score(y_test, lr_pred))
print("F1 Score:", f1_score(y_test, lr_pred))
```

```
Logistic Regression Performance:
Accuracy: 0.8836107303700803
Precision: 0.5400943396226415
Recall: 0.02558659217877095
F1 Score: 0.04885854491145722
```

Figure 8: Logistic Regression Model

3.2 Decision Tree Classifier

The Decision Tree classifier creates rule-based frameworks that divide the dataset according to thresholds of features in order to classify borrowers.

The model had an accuracy of 0.8008, precision of 0.1902 and a recall of 0.2164. The Decision Tree model had the capability to identify more cases of default than Logistic Regression, which is indicated by its higher recall.

The F1-score value of 0.2025 suggests that the precision and recall have a more balanced performance than Logistic Regression. Decision Trees have the capability of reflecting nonlinear associations between the traits of borrowers and loan default risk..

```python
# Decision Tree Classifier
dt_model = DecisionTreeClassifier(random_state=88)
dt_model.fit(X_train, y_train)
dt_pred = dt_model.predict(X_test)

print("\nDecision Tree Classifier Performance:")
print("Accuracy:", accuracy_score(y_test, dt_pred))
print("Precision:", precision_score(y_test, dt_pred))
print("Recall:", recall_score(y_test, dt_pred))
print("F1 Score:", f1_score(y_test, dt_pred))
```

```
Decision Tree Classifier Performance:
Accuracy: 0.8007962926701913
Precision: 0.19020031421838177
Recall: 0.2164245810055866
F1 Score: 0.20246681300303127
```

Figure 9: Decision Tree Model

3.3 Random Forest Classifier

The basic model was the Logistic Regression model that predicts default of loans. It provided an accuracy of 88.36 implying that majority of the predictions were right. The recall was however very low (2.56%), which indicated that the model was unable to detect a large number of borrowers that ended up defaulting. The F1-score stood at 4.89% and this is also poor performance in loan default detection.

All in all, it is a simple Logistic Regression, and it is not very effective in identifying the most at-risk borrowers.

```
# Random Forest Classifier
rf_model = RandomForestClassifier(n_estimators=100, random_state=88)
rf_model.fit(X_train, y_train)
rf_pred = rf_model.predict(X_test)

print("\nRandom Forest Classifier Performance:")
print("Accuracy:", accuracy_score(y_test, rf_pred))
print("Precision:", precision_score(y_test, rf_pred))
print("Recall:", recall_score(y_test, rf_pred))
print("F1 Score:", f1_score(y_test, rf_pred))
```

```
Random Forest Classifier Performance:
Accuracy: 0.8836237843482802
Precision: 0.5201845444059977
Recall: 0.050391061452513965
F1 Score: 0.09188143017215035
```

Figure 10: Random Forest Model

4 Reflection

Challenges Faced

- Class imbalance: There was a vast number of fewer cases of default than non-default in the dataset.
- Relationships of features: There were relationships between some financial variables that may affect the interpretation of the model.
- Size of a dataset: Due to the size of a dataset of over 250,000 records, it was necessary to process the data efficiently.

Key Learnings

- The relationship between age and Default is the most negative.
- The positive relationship between loan default and interest rate is the greatest.
- Increase in income and long working years minimize default risk to a small extent.

Table 1: Model Performance Comparison

Metric	Logistic Regression	Decision Tree	Random Forest	Best Performer
Accuracy	0.8836	0.8008	0.8836	Logistic Regression / Random Forest
Precision	0.5401	0.1902	0.5202	Logistic Regression
Recall	0.0256	0.2164	0.0504	Decision Tree
F1 Score	0.0489	0.2025	0.0919	Decision Tree

5 Conclusion

Whereas the accuracy of Logistic Regression and the Random Forest was the best, the Decision Tree classifier was the most suitable with higher recall and F1-score, which imply that it has a greater potential to identify borrowers who are most likely to default. Hence, Decision Tree model is more balanced when it comes to identification of high-risk borrowers.

6 References

[1] F. Chollet, "Deep Learning with Python.," Manning Publications., 2018.

[2] A. Géron, "Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow.," O'Reilly Media., 2023.

[3] W. McKinney, "Python for Data Analysis.," O'Reilly Media., 2022.

[4] J. VanderPlas, "Python Data Science Handbook.," O'Reilly Media., 2017.

Data set link : https://www.kaggle.com/datasets/nikhil1e9/loan-default

GitHub repository link : https://github.com/RamarThivagar/datascience_assigment_14520/blob/main